

THABOLEZWE MABANDLA

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SUMMARY

Graduate student in Engineering Artificial Intelligence at Carnegie Mellon University Africa, focused on robotics, embodied AI, and machine learning. Hands-on experience developing robotic systems with ROS, ROS 2, C++, Python, Gazebo, and reinforcement learning, including robot control, autonomous navigation, and adversarial robot-learning research. Complemented by industry experience building production ML infrastructure, data pipelines, and LLM systems, with current work extending into vision-language-action models and robust robotic manipulation.

EDUCATION

Masters of Science in Engineering AI

August 2025 - May 2027

Carnegie Mellon University, Africa

Concentration: Robotics and Embodied AI

BACHELOR OF SCIENCE HONOURS DEGREE IN AI & MACHINE LEARNING

August 2020 - June 2024

University of Zimbabwe, Harare

Graduated with First Class Honors

Awarded the University of Zimbabwe Book Prize

Awarded the Best Undergraduate Researcher and Innovator in the Faculty of Computer Engineering

RESEARCH WORK

Agent-Augmented VLA for Training-Free Policy Adaptation

September 2026 - Present

- Designing an agentic supervision harness that monitors a frozen Vision-Language-Action (VLA) policy at runtime, detects execution failures, and applies corrective actions without retraining or collecting new demonstrations.
- Building and evaluating a LIBERO-based proof of concept on pick-and-place manipulation tasks, comparing baseline policy success against supervised execution with failure detection, intervention logging, and correction memory.
- Developing a pathway for sim-to-real transfer to bimanual YAM robots, with intermediate validation on T-shirt folding and final generalization evaluation on towel folding using task success rate across repeated trials.

Adversarial RL for Indirect Robot Control

May 2026 - Present

- Developed a ROS 2+Gazebo reinforcement learning framework where an embodied “parasite” agent manipulates an autonomous iRobot Create 2 indirectly through virtual bumper interventions rather than direct motor control.
- Modeled parasite actions as temporally extended options in a semi-Markov decision process (SMDP), incorporating variable action durations and potential-based reward shaping over goal distance and heading.
- Currently training and evaluating PPO policies that learn sparse, goal-directed interventions under partial observability.

EXPERIENCE

Graduate Teaching Assistant

August 2026 - Present

Carnegie Mellon University Africa, Kigali

- Facilitate weekly recitations and office hours for graduate students, clarifying theoretical foundations.
- Guide students through computational simulations and numerical modeling in Python
- Design assessment rubrics, evaluate analytical problem sets and exams, and provide targeted technical feedback.

Selected Participant

August 2026 - August 2026

Cornell, Maryland, Max Planck Pre-doctoral Research School, Saarbrücken, Germany

- Engaged in advanced computer science research lectures and discussions spanning emerging research areas, research methodology, and academic and industrial research careers.
- Participated in faculty-led sessions and PhD-focused panels covering research problem formulation, graduate research pathways, and doctoral admissions.

Graduate Research Assistant

June 2026 - August 2026

Carnegie Mellon University Africa, Kigali

- Conducted research on indirect "parasite-host" robotic control between two non-communicating robotic agents with different goal programming.
- Trained a hierarchical RL agent (lynxmotion manipulator arm) to steer a Roomba base purely via tactile bumper interactions.
- Integrated Nav2 and Gazebo Harmonic simulations using SMDP macro-actions to model physical actuator latency and eliminate command jitter.

Backend Developer/ Machine Learning Engineer

March 2025 - August 2025

CyberAlliance, Raleigh, North Carolina

- Engineered scalable data pipelines with Apache Airflow, Kafka, Redis, and Postgres, improving workflow automation and reliability on distributed systems.
- Fine-tuned LLMs (HuggingFace Transformers, BitsandBytes) using MLFlow and AWS, optimizing inference efficiency and reducing training costs.
- Implemented RAG pipelines with Weaviate, Elastic Stack, and Sentence Transformers, enhancing retrieval accuracy and contextual search for enterprise clients.
- Deployed a MISP threat intelligence server for real-time ingestion and sharing of cyber threat data across teams.
- Integrated multiple CTI clients (NVD NIST, Abuse.ch, AlienVault OTX, MITRE.org), streamlining security workflows and strengthening proactive defense capabilities.

AI Software Developer

October 2024 - March 2025

Rubiem Innovations, Harare, Harare

- Led the development of a Local LLM agent prototype for Zimpapers (Zimbabwe's largest media house), enhancing journalists' article-writing workflow with enhanced retrieval from internal knowledge bases and integrated Google Search capabilities.
- Designed and deployed LLM-based applications that automate customer support and content generation for enterprise clients.
- Assisted in the process of providing AI strategy guidance for banks and telecoms, aligning organizational needs with scalable AI adoption.
- Translated technical insights into actionable recommendations for non-technical stakeholders, while delivering training and ongoing client support.

Research Scientist

January 2024 - June 2024

Zimbabwe Centre for High Performance Computing, Harare

- Conducted NLP research for developing language technologies for Zimbabwe's two main indigenous languages.
- Developed and deployed a novel Shona spelling correction LLM, the first of its kind for one of Zimbabwe's main native languages.
- Built machine learning workflows on HPC clusters, accelerating large-scale model training and experimentation.

IT Operations Intern

September 2022 - August 2023

Old Mutual & CABS, Harare

- Monitored and optimized server uptime, network connectivity, and application performance, ensuring continuous availability of financial systems.
- Partnered with software teams to deploy and update applications, reducing deployment errors and improving turnaround time.

SKILLS

- **Programming & Scripting:** (advanced) Python, (basic) C++, JavaScript, SQL, Bash.
- **Machine Learning & AI:** RL, Imitation Learning, PyTorch, ROS 2, Nav 2, MoveIt 2, Gazebo Sim
- **Data Engineering & Databases:** Apache Airflow, Kafka, Redis, Postgres, Elastic Stack, Weaviate, Docker, Ubuntu Server
- **Tools & Platforms:** Git, MLFlow, BitsandBytes, Google Cloud, wandb, Canva.